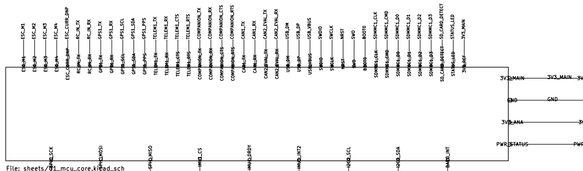
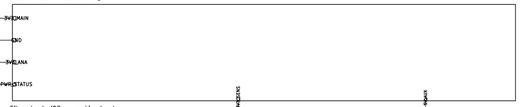
STM32H743 core

Stage07 interfaces captured; recommendations OPEN UNTESTED. No Stage08 authority.



File: sheets/01_mcu_core.kicad_sch

Power and monitoring



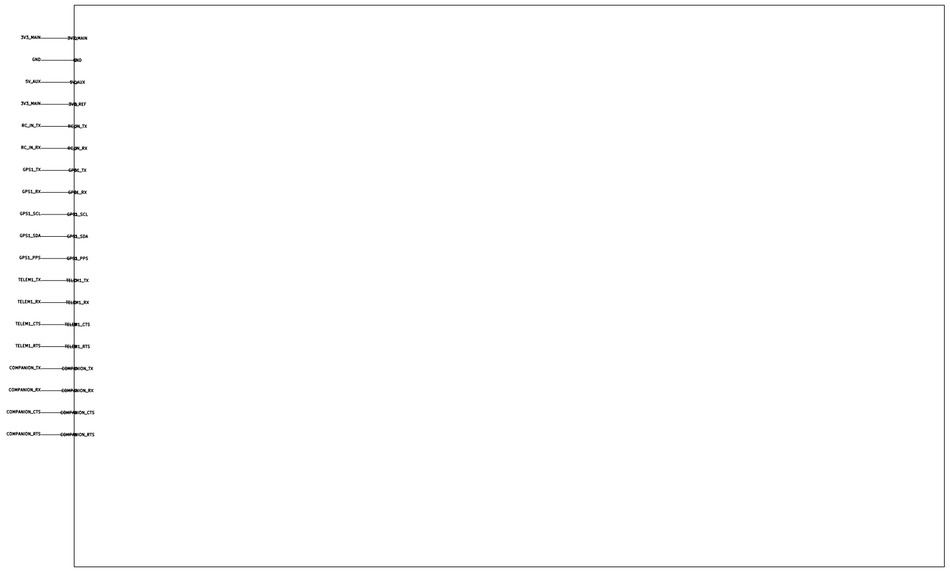
File: sheets/02_power.kicad_sch

Local sensors



File: sheets/03_sensors.kicad_sch

External UART and I2C ports



File: sheets/05_external_ports.kicad_sch

ESC signal boundary



File: sheets/04_esc.kicad_sch

CAN interfaces



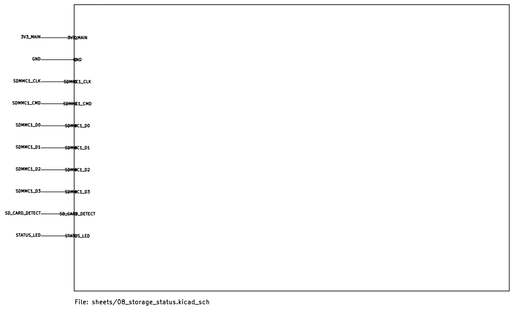
File: sheets/06_can.kicad_sch

USB and debug



File: sheets/07_usb_debug.kicad_sch

Storage and status



File: sheets/08_storage_status.kicad_sch

ENGINEERING VALIDATION EVT1

Desan Vasu

Sheet: /

File: FC_REV_A.kicad_sch

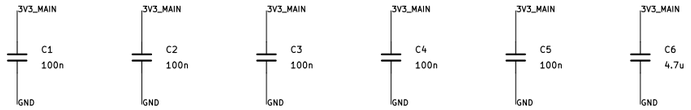
Title: Flight Controller V1

Size: A2	Date: 2026-09-14	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/9

FC FC REV A | STM32H743VIT6 CORE | STAGE 04

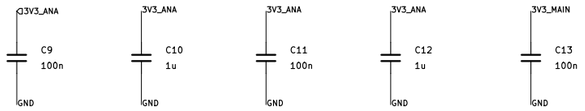
OPEN — UNTESTED. Schematic capture only; all production footprints and layout await Stage 08/09.

DIGITAL SUPPLY | 5 VDD/VSS PAIRS



C1..C5: pairs 11/10, 27/26, 50/49, 75/74, 100/99.

ANALOG / REFERENCE / VBAT



C9/C10: VDDA 21. C11/C12: VREF+ 20. C13: VBAT 6.

VSSA -> common GND. VREFBUF: ENVR=0, HIZ=1.

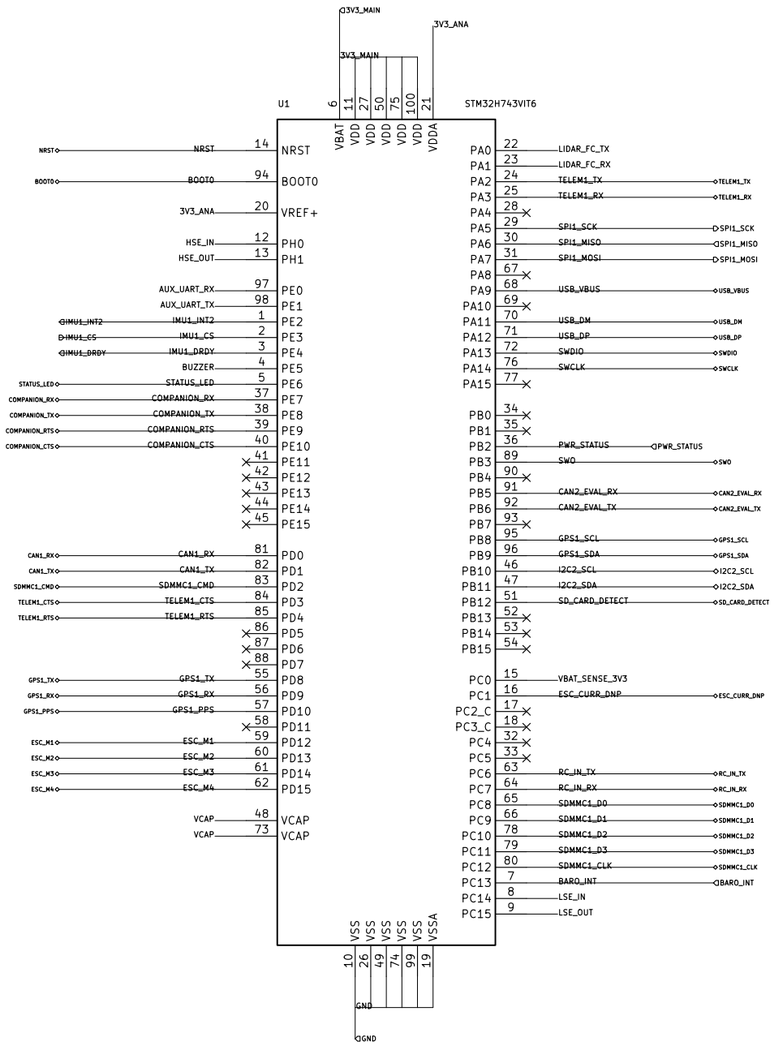
3V3_ANA generation/sequencing: Stage 05. No ground island.

INTERNAL LDO | ISOLATED VCAP

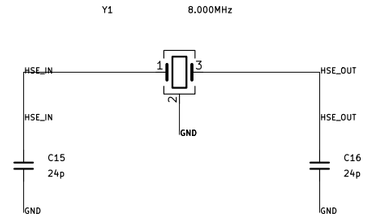


Pins 48 + 73 tied: AN4938 Rev 7, section 2.2.

One capacitor at each pin. No load, test point or bypass.



HSE | 8.000 MHz

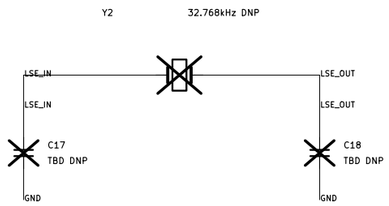


RECOMMENDATION: ABM3B-8.000MHZ-B2-T: CL=18pF.

C15=C16=24pF C0G 1%; Cs=6pF ASSUMPTION.

No oscillator test points. Startup/drive/layout remain OPEN.

LSE | DNP / RESERVED



Y2/C17/C18 DNP. No MPN or load values frozen.

USB FS | MCU-SIDE BOUNDARY

PA9 / pin 68 = USB_VBUS (sense only).

PA11 / 70 = USB_DM; PA12 / 71 = USB_DP.

No VBUS-to-FC rail path. No connector/CC/ESD here.

Stage 07: protect/condition VBUS for all power states;

preserve ROM bootloader behavior and no backfeed.



PWR_FLAG = required Stage 05 driven boundary; no regulator captured.

Unused pins carry explicit NC marks. PC2_C/PC3_C are analog-only.

All peripheral labels are reservations; external circuits are NOT captured.

CAN2_EVAL remains DNP/reserved. ESC_CURR_DNP is informational only.

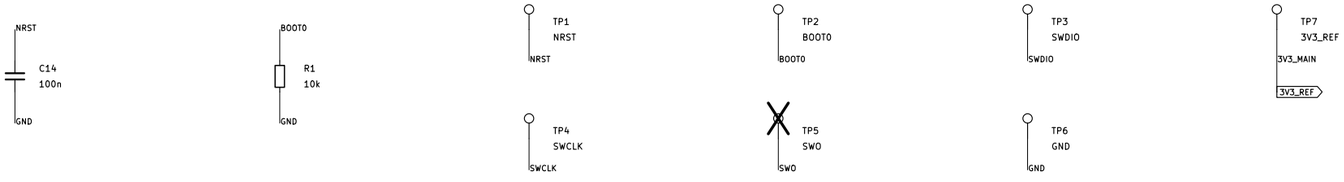
LSE is DNP; PC13/BARO_INT conflict requires errata review before LSE population.

3V3_REF is a direct voltage-sense alias of 3V3_MAIN, not a power input.

NRST/BOOT0 recovery: probe using common GND; apply only 3.3 V.

Stage 13: verify boot options, silicon revision, clocks, reset and USB.

RESET / BOOT / DEBUG ACCESS



OPEN — UNTESTED; NOT FOR FABRICATION

FC Drone Project

Sheet: /STM32H743 core/

File: 01_mcu_core.kicad_sch

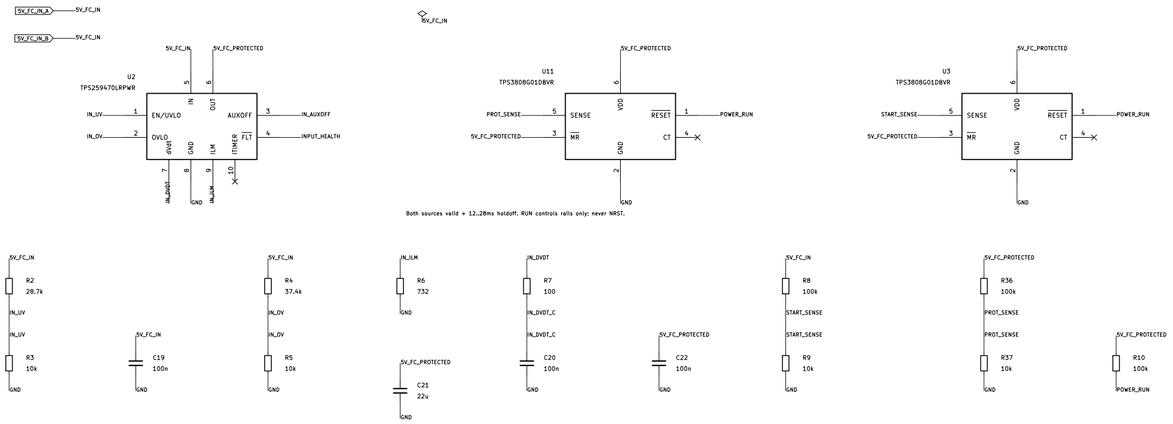
Title: FC FC Rev A — STM32H743 core

Size: A3	Date: 2026-09-08	**Rev: A-S04-01**
KiCad E.D.A. 10.0.6		Id: 2/9

FC FC REV A | POWER TREE / PROTECTION / MONITORING | R18

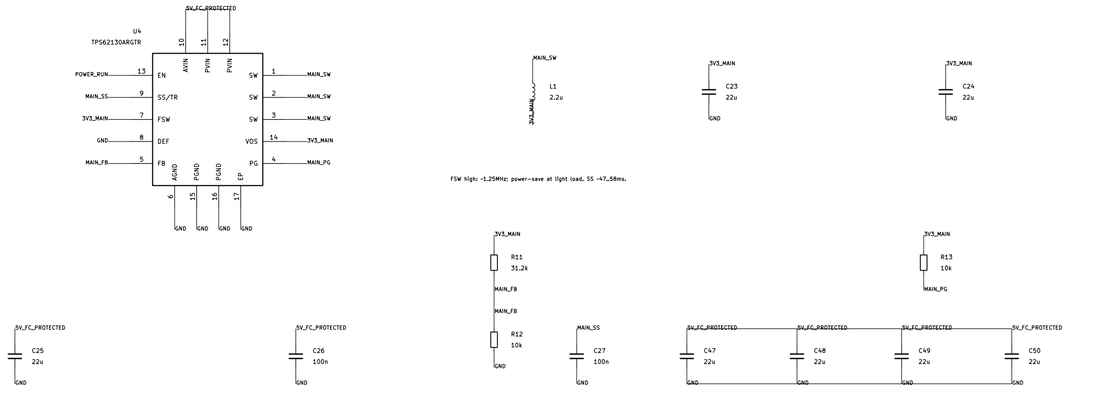
RECOMMENDATION CAPTURE — OPEN / UNTESTED. Physical SOA, stability, EMI, thermal and transients: coupon / Stages 11–13.

01 | POWER1 PROTECTION + HARDWARE STARTUP



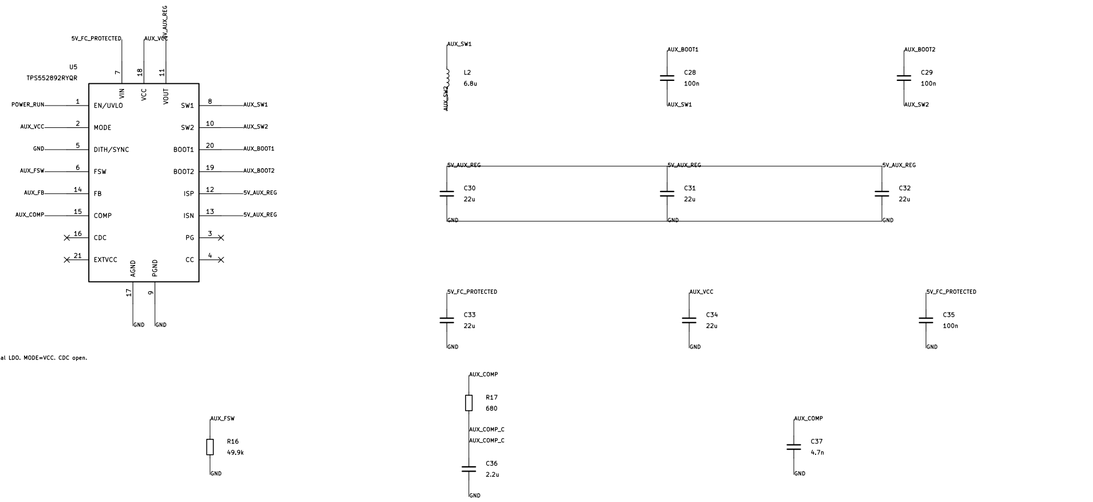
Both sources valid + 12..28ms holdoff. RUN controls rails only; never NRST.

02 | 3V3_MAIN — 3A BUCK / 1.5A CONTINUOUS



FSW high: ~1.25MHz; power-save at light load. SS ~47.5ms.

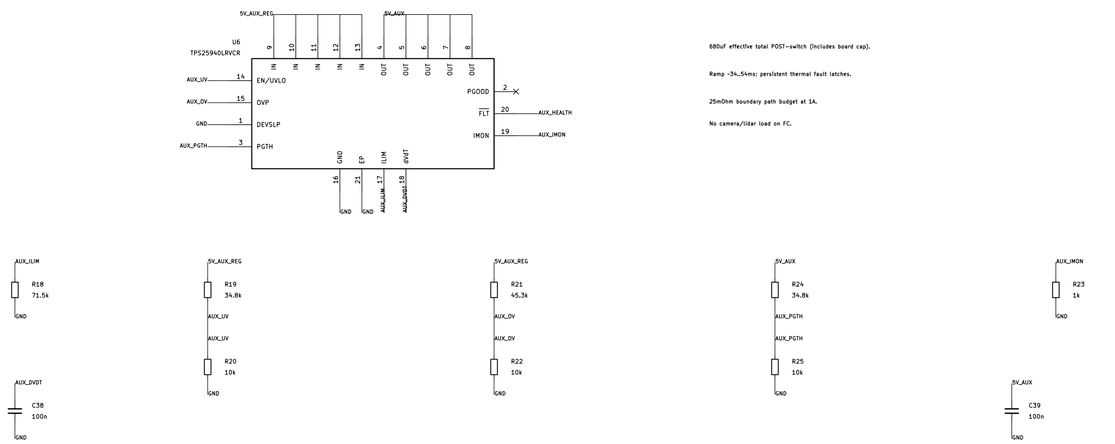
03 | AUX BUCK-BOOST — LOCAL FEEDBACK BEFORE SWITCH



Pin21 EXTVCC left open: internal LDO. MODE=VCC. CDC open.

04 | 5V_AUX — 1A TOTAL / LATCH-OFF LIMITER

AUX load: OpenRX <=0.5A; all other modules share <=0.5A.



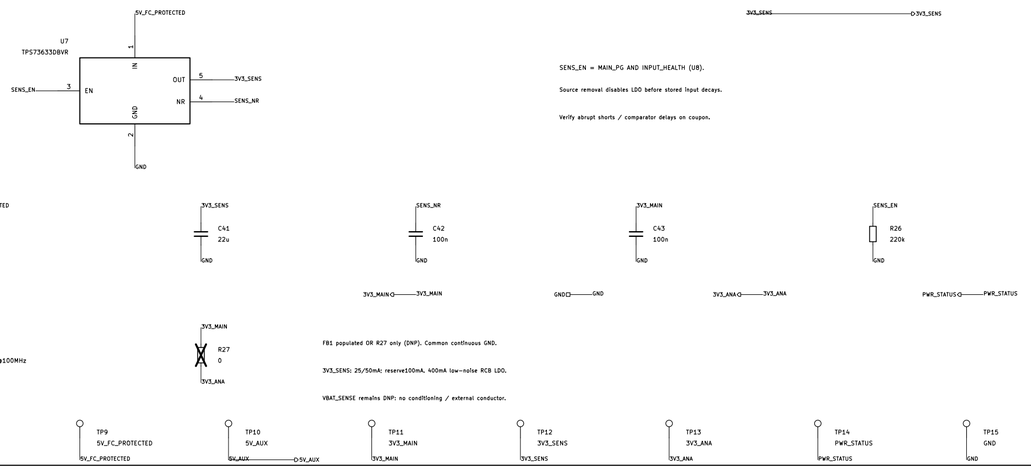
680uF effective total POST-switch (includes board cap).

Ramp ~34..54ms: persistent thermal fault latches.

25mOhm boundary path budget at 1A.

No camera/lidar load on FC.

05 | QUIET SENSOR / ANALOG RAILS + ACCESS



SENS_EN = MAIN_PG AND INPUT_HEALTH (U8).

Source removal disables LDO before stored input decays.

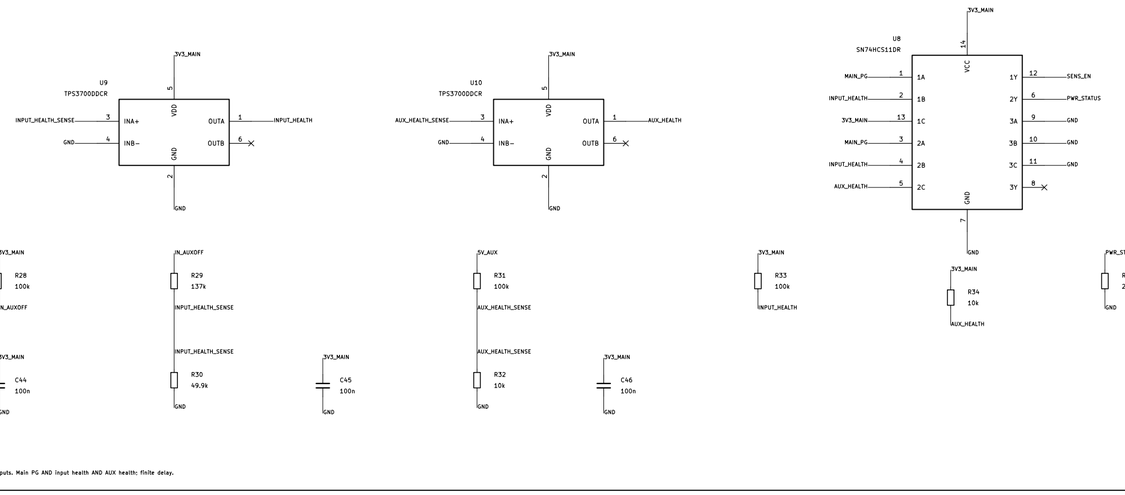
Verify abrupt shorts / comparator delays on coupon.

FB1 populated OR R27 only (DNP). Common continuous GND.

3V3_SENS: 25/50mA; reserve100mA. 400mA low-noise RCB LDO.

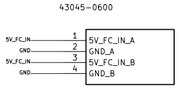
VBAT_SENSE remains DNP: no conditioning / external conductor.

06 | INFORMATIONAL FAIL-LOW PWR_STATUS



Schmitt inputs. Main PG AND input health AND AUX health: finite delay.

J_POWER1
43045-0600



POWER1: physical 1=5V A, 2=GND A, 3=5V B, 4=GND B.
5/6: NO cable terminals; header tails in NPTH clearance holes: NO copper.
43025-0600 housing; 43030-0038 terminals, 18 AWG.
RECOMMENDATION / fit, crimp, strain relief and thermal validation OPEN.

RECOMMENDATION — OPEN UNTESTED — NOT FOR FABRICATION

FC Drone Project

Sheet: /Power and monitoring/
File: 02_power.kicad_sch

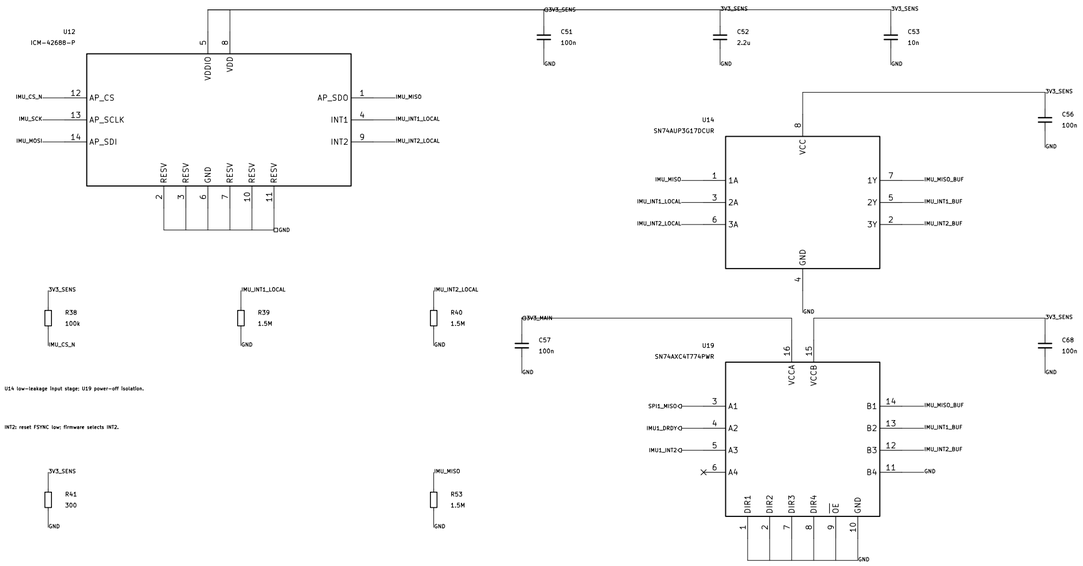
Title: FC FC Rev A — Power tree

Size: User	Date: 2026-09-09	**Rev: A-S05-R18**
KiCad E.D.A. 10.0.6		Id: 3/9

FC FC REV A | ONBOARD IMU / BAROMETER | STAGE 06

RECOMMENDATION / OPEN PHYSICAL VALIDATION — NO PCB OR FABRICATION RELEASE

01 | ICM-42688-P — DIRECT QUIET RAIL / SPI1



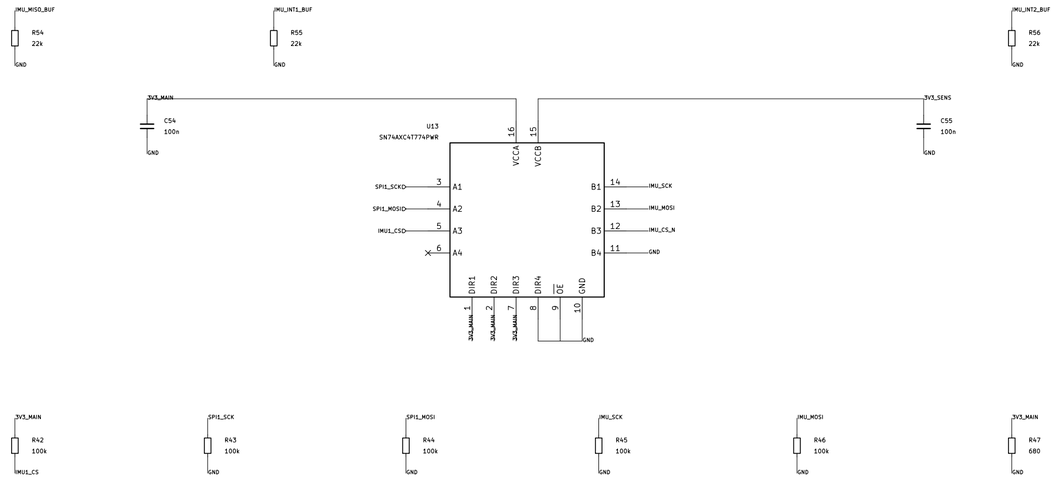
U14 low-leakage input stage; U19 power-off isolation.

INT2: reset FSYNC low; firmware selects INT2.

RESV 2/3/10/11 grounded (allowed); RESV7 MUST GND. No EP or REGOUT pin.

C51+C52 beside VDDR; C53 beside VDDIO5. Direct 3V3_SENS.

02 | SPI / PARTIAL-POWER BOUNDARY



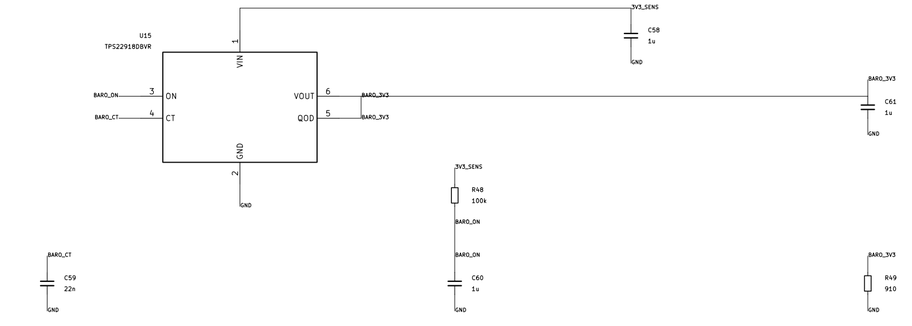
A = MCU / MAIN; B = IMU / SENS. Ch1..3 outbound; Ch4 unused. OE grounded.

Power-off isolation / Ioff rated parts. Bleeders bound leakage; no external connector.

Bring-up: SPI mode 0, 1MHz; CS high before AF selection; verify timing before 24MHz.

Qualify direct 3V3_SENS monotonic 1..3ms and sensor load <=25mA / <=50mA.

03 | BMP LOCAL SLEW — CONDITIONAL / RAMP VALIDATION OPEN



CT22nF: ~40ms TYP 10-90%; equation is NOT a guaranteed minimum.

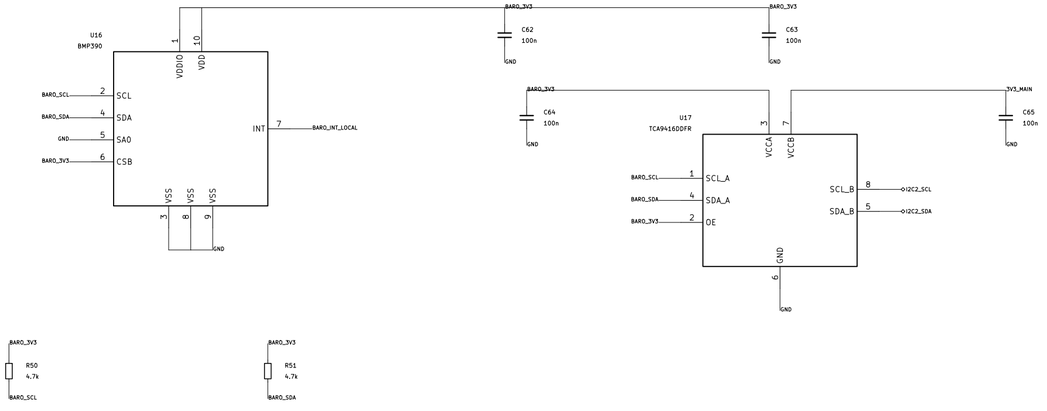
ON RC delays turn-on after SENS settles; QOD tied output: 910R bleeder.

OPEN: >=10ms Bosch ramp across PVT, restart, rail decay and partial power.

No bypass link. Do not energize BMP390 until coupon waveform qualifies.

Local support only; Stage05 regulator / EN / compensation unchanged.

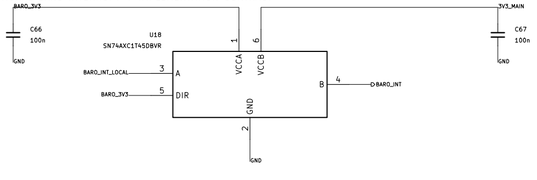
04 | BMP390 — I2C2 / 0x76 / PROTECTED PRESSURE VENT



CSB strapped VDDIO per Fig25: SDO=GND gives 0x76. No EP.

TCA9416: internal 10k each side + 4.7k on A; OE=A rail. Start at100kHz.

05 | OPTIONAL BARO INTERRUPT + IMPLEMENTATION CONTRACT



Vent: keep open; protect from propwash, dust, liquids, adhesive / coating and thermal bias.

Continuous GND. Short local SPI; isolate from switch nodes, vibration, mounting stress / heat.

Firmware: wait stable supplies, reset / check WHO_AM_I and CHIP_ID; configure IRQ after bus.

Exact footprints / vent geometry / assembly / EMC / flight qualification remain OPEN.

RECOMMENDATION / OPEN PHYSICAL VALIDATION — NOT FOR FABRICATION

FC Drone Project

Sheet: /Local sensors/
File: 03_sensors.kicad_sch

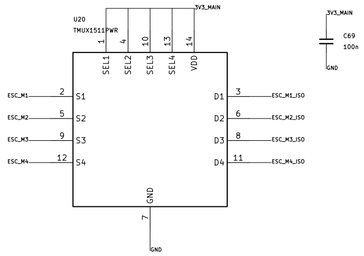
Title: FC FC Rev A — IMU / barometer

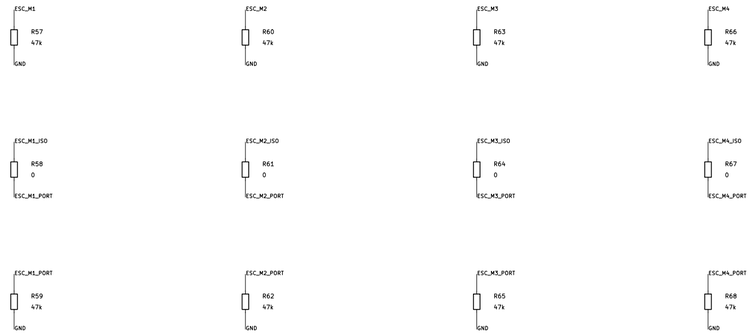
Size: A2	Date: 2026-09-09	**Rev: A-S06-01**
KiCad E.D.A. 10.0.6		Id: 4/9

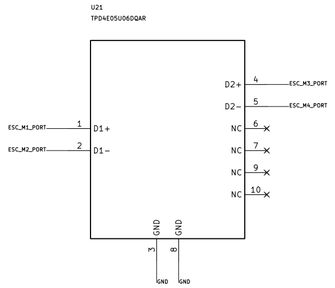
ESC: sparse bench boundary — Stage07 recommendation / OPEN UNTESTED

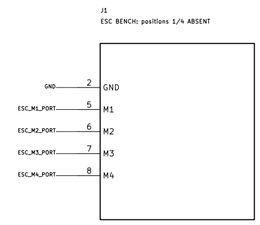
3.3V logic. No external 5V direct to MCU. No PCB / footprint release / fabrication authority.

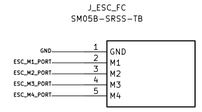
Motor signals / bidirectional DShot









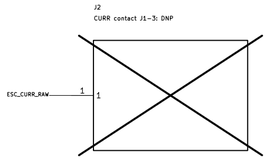


J1 has ONLY 2,5,6,7,8. No 1/4 pins, pads, wires or nets. Controlled interposer <=150mm; flight harness OPEN.

OR series default; evaluate 22R only after bidirectional waveform test. Both sides 47k low. ESC firmware FT pulls OFF.

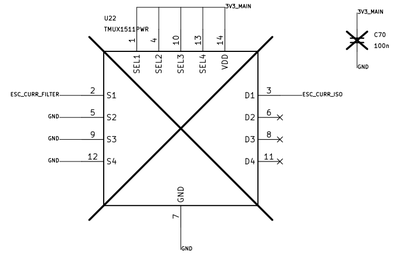
Physical FC connector: 1=GND, 2=M1, 3=M2, 4=M3, 5=M4.
Sparse logical J1 retained; J2 current-sense remains DNP / footprintless.
NO ESC battery, power or current-sense conductor on this harness.
RECOMMENDATION / unpowered continuity and powered bench validation OPEN.

Optional current / NOT a safety input

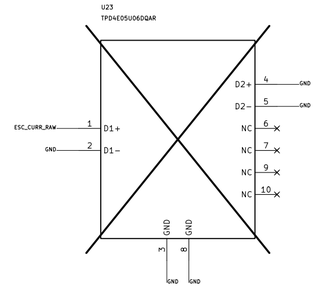












Entire optional current branch DNP including contact, divider, switch, bypass and protection. PC1 ADC disabled.

Divider recommendation only; verify upstream CURR range/calibration before any population. No safety dependency.

~~3V3_MAIN~~ 3V3_MAIN
~~GND~~ GND
~~ESC_M1~~ ESC_M1
~~ESC_M2~~ ESC_M2
~~ESC_M3~~ ESC_M3
~~ESC_M4~~ ESC_M4
~~ESC_CURR_ISO~~ ESC_CURR_DNP

RECOMMENDATION — OPEN UNTESTED — NOT FOR FABRICATION

FC Drone Project

Sheet: /ESC signal boundary/
File: 04_esc.kicad_sch

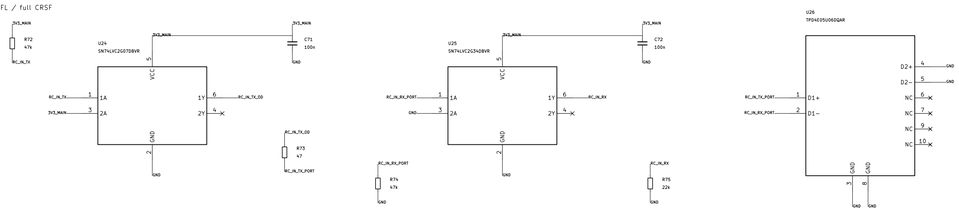
Title: FC FC Rev A Stage07 — esc

Size: A2	Date: 2026-09-09	**Rev: A-S07-01**
KiCad E.D.A. 10.0.6		Id: 5/9

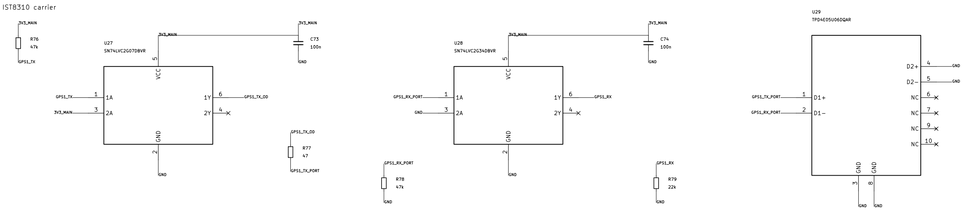
External UART /I2C contracts + Stage07 recommendation / OPEN UNTESTED

3.3V logic. No external 5V direct to MCU. No PCB / footprint release / fabrication authority.

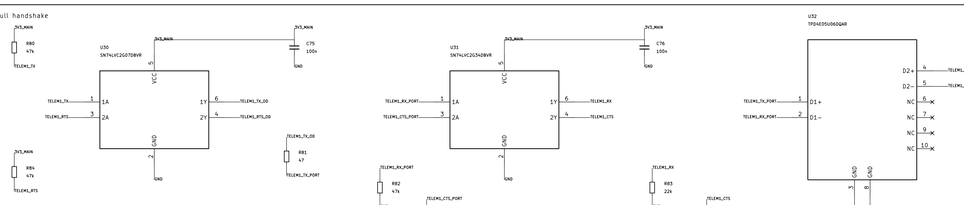
RC_IN — OpenRX Lite-UFL / full CRSF



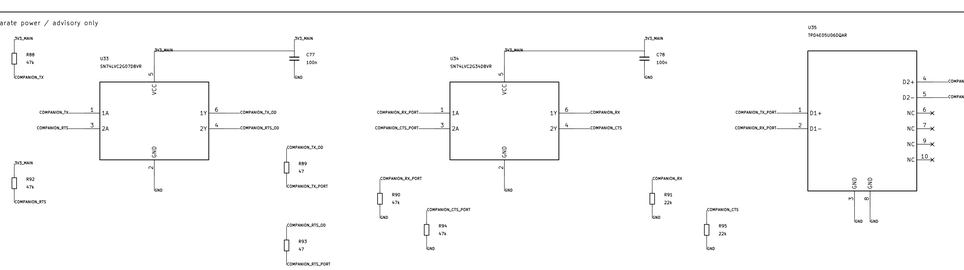
GPS1 — MAX-M10S + IST8310 carrier



TELEM1 — RFD900x / full handshake



COMPANION — CM5 separate power / advisory only



3V3_MAIN — 3V3_MAIN

GND — GND

5V_MAIN — 5V_AUX

5V0_REF — 3V3_MAIN

RC_IN_TX — RC_IN_TX

RC_IN_RX — RC_IN_RX

GPS1_TX — GPS1_TX

GPS1_RX — GPS1_RX

GPS1_SCL — GPS1_SCL

GPS1_SDA — GPS1_SDA

GPS1_PPS — GPS1_PPS

TELEM1_TX — TELEM1_TX

TELEM1_RX — TELEM1_RX

TELEM1_CTS — TELEM1_CTS

TELEM1_RTS — TELEM1_RTS

COMPANION_TX — COMPANION_TX

COMPANION_RX — COMPANION_RX

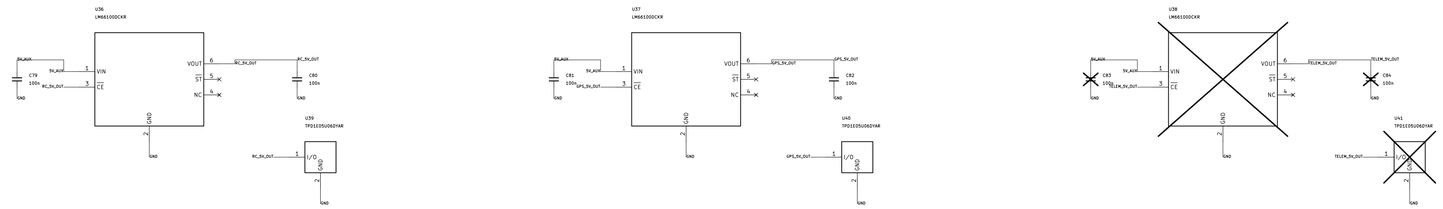
COMPANION_CTS — COMPANION_CTS

COMPANION_RTS — COMPANION_RTS

REMOTE REQUIREMENT: 2.2k pull-up on each FC TX / RTS at remote OWN 3.3V. No FC pull-up on exposed outputs.

Remote TX / RTS >=2.0V high, <=0.8V low. Remote adapter is required; this is not an unqualified bare-cable contract.

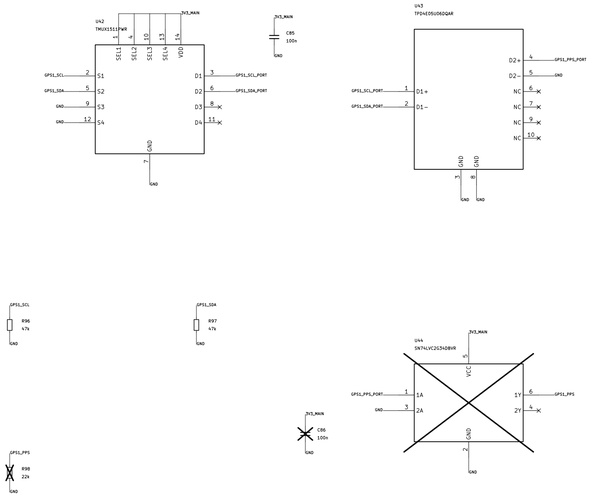
Port power reverse blocking / shared AUX budget



AUX <=1A total / <=680uF total post-switch. RC reserve 0.5A, GPS allocation 0.15A, TELEM power DNP.

Each powered module must obey Stage05 ramp/startup envelope. No independent 5V backfeed through diode paths.

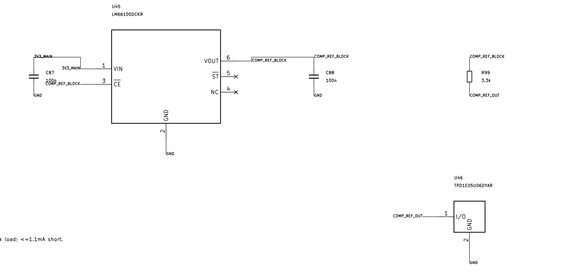
GPS open-drain I2C / optional PPS



Remote owns 2.2k pulls to remote 3.3V.

<=400kHz, <=300mm, total C <=100pF.

Companion reference only / inactive resources



Sense only >=100k load: <=1.1mA short.

No FC power to CM5, camera or lidar.

LIDAR_FC retired: PA0/PA1 reserved.

AUX_UART PE1/PE0 disabled / unassigned.

BOOT receiver service: remote-side only.

CRSF 420k; GNSS UART 115200.

TELEM initial 57600; 921600 needs test.

CM5 UART7: explicit UART/flow config.

RECOMMENDATION — OPEN UNTESTED — NOT FOR FABRICATION

FC Drone Project

Sheet: /External UART and I2C ports/

File: 05_external_ports.kicad_sch

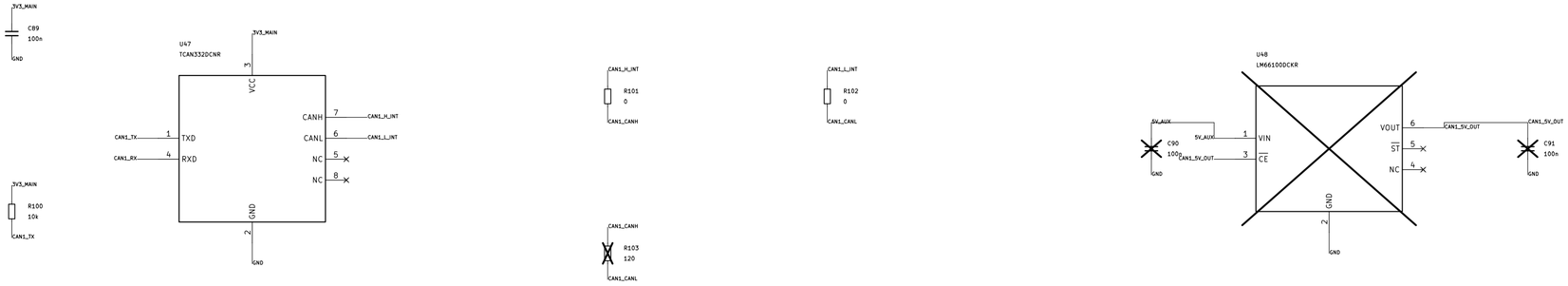
Title: FC FC Rev A Stage07 — external_ports

Size: A1	Date: 2026-09-09	**Rev: A-S07-01**
KiCad E.D.A. 10.0.6		Id: 6/9

Classic CAN — Stage07 recommendation / OPEN2UNTESTED

3.3V logic. No external 5V direct to MCU. No PCB / footprint release / fabrication authority.

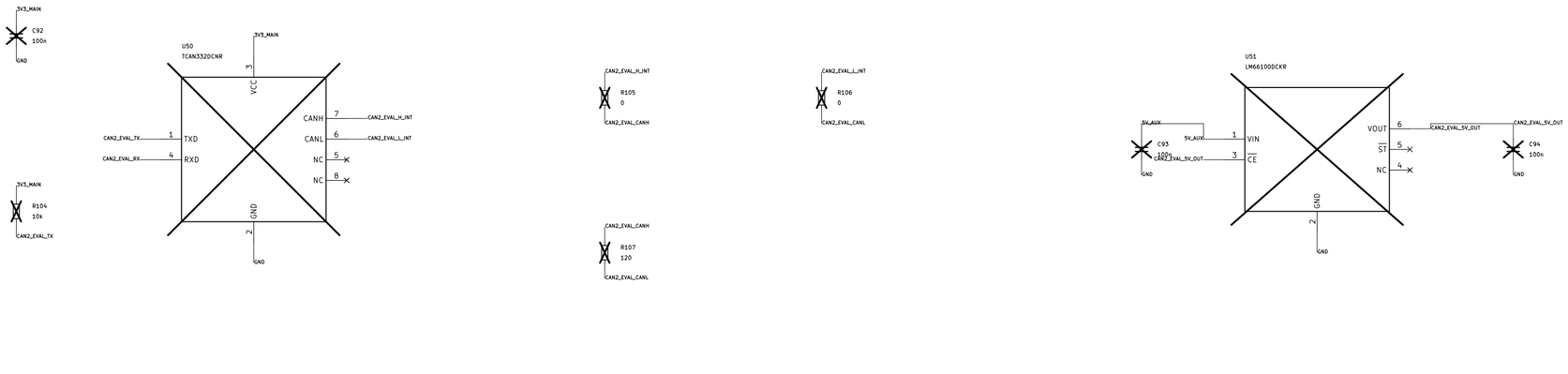
CAN1 — populated 1Mbit/s <=1m



Integrated bus ESD: +/−12kV IEC contact. OR bypasses: CMC option requires reviewed EMI change.

120R termination DNP; fit ONLY at bus end. 5V output DNP. Nominal common mode +/−12V; abs +/−14V.

CAN2_EVAL — ENTIRE PATH DNP



Integrated bus ESD: +/−12kV IEC contact. OR bypasses: CMC option requires reviewed EMI change.

120R termination DNP; fit ONLY at bus end. 5V output DNP. Nominal common mode +/−12V; abs +/−14V.

~~3V3_MAIN~~ 3V3_MAIN

~~CAN1_TX~~ CAN1_TX

~~CAN2_EVAL_RX~~ CAN2_EVAL_RX

~~GND~~ GND

~~CAN1_RX~~ CAN1_RX

~~5V_AUX~~ 5V_AUX

~~CAN2_EVAL_TX~~ CAN2_EVAL_TX

RECOMMENDATION — OPEN UNTESTED — NOT FOR FABRICATION

FC Drone Project

Sheet: /CAN interfaces/

File: 06_can.kicad_sch

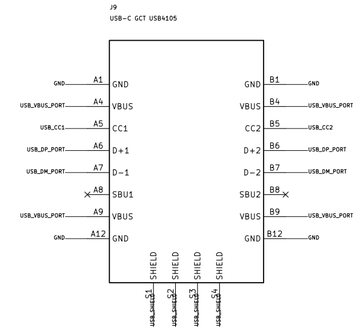
Title: FC FC Rev A Stage07 — can

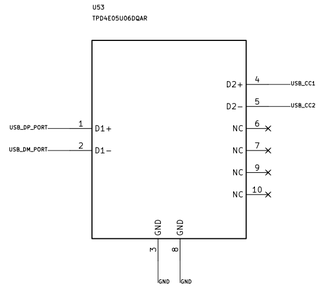
Size: A2	Date: 2026-09-09	**Rev: A-S07-01**
KiCad E.D.A. 10.0.6		Id: 7/9

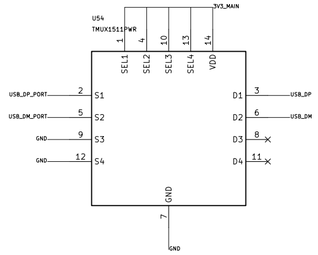
USB data-only / SWD recovery — Stage07 recommendation / OPEN UNTESTED

3.3V logic. No external 5V direct to MCU. No PCB / footprint release / fabrication authority.

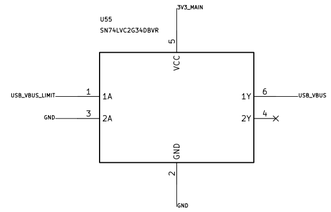
USB-C self-powered FS / VBUS sense only









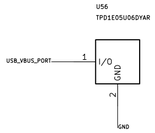












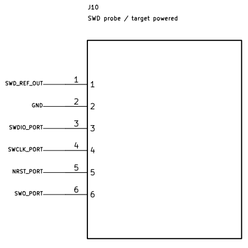


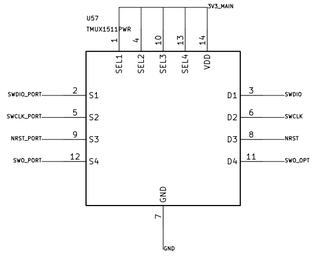




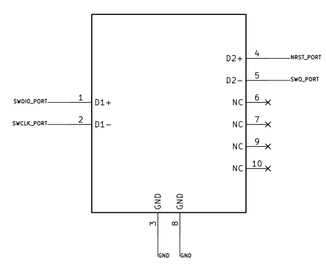
No VBUS-to-rail path. Ground-referenced TVS replaces USBLC steering array. No PD source/sink controller.

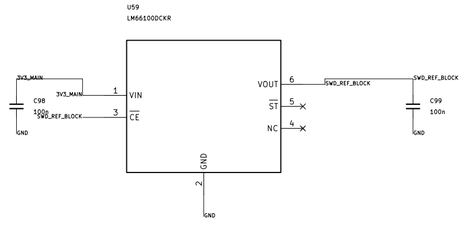
SWD and boot recovery / reference only





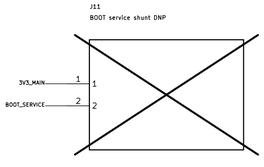








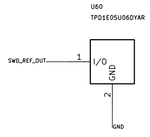






680R MAIN bleed: <=5.2mA





No reset button / new supervisor. Existing NRST network retained. BOOT shunt local only; USB DFU firmware validation OPEN.

~~3V3_MAIN~~ 3V3_MAIN
~~3V3_REF~~ 3V3_MAIN
~~GND~~ GND

~~USB_DM~~ USB_DM
~~USB_DP~~ USB_DP
~~USB_VBUS~~ USB_VBUS

~~SWDIO~~ SWDIO
~~SWCLK~~ SWCLK
~~NRST~~ NRST

~~SWO~~ SWO
~~BOOT0~~ BOOT0

RECOMMENDATION — OPEN UNTESTED — NOT FOR FABRICATION

FC Drone Project

Sheet: /USB and debug/
File: 07_usb_debug.kicad_sch

Title: FC FC Rev A Stage07 — usb_debug

Size: A2	Date: 2026-09-09	**Rev: A-S07-01**
KiCad E.D.A. 10.0.6		Id: 8/9

Removable microSD / optional status — Stage07 recommendation / OPEN UNTESTED

3.3V logic. No external 5V direct to MCU. No PCB / footprint release / fabrication authority.

SDMMC1 4-bit / 3V3_MAIN only

SDMMC1_CLK
R118
22
SD_CLK_PORT

SDMMC1_CMD
R119
0
SD_CMD_PORT

SDMMC1_D0
R121
0
SD_D0_PORT

SDMMC1_D1
R123
0
SD_D1_PORT

SDMMC1_D2
R125
0
SD_D2_PORT

SDMMC1_D3
R127
0
SD_D3_PORT

3V3_MAIN
R120
47k
SD_CMD_PORT

3V3_MAIN
R122
47k
SD_D0_PORT

3V3_MAIN
R124
47k
SD_D1_PORT

3V3_MAIN
R126
47k
SD_D2_PORT

3V3_MAIN
R128
47k
SD_D3_PORT

J12
microSD GCT MEM2075

SD_D2_PORT	1	DAT2	
SD_D3_PORT	2	CD/DAT3	
SD_CMD_PORT	3	CMD	
3V3_MAIN	4	VDD	
SD_CLK_PORT	5	CLK	
GND	6	VSS	
SD_D0_PORT	7	DAT0	
SD_D1_PORT	8	DAT1	
		DETECT	CD → SD_CARD_DETECT
		SHELL	SH → GND

U61
TPD4E05U06DQAR

SD_CLK_PORT	1	D1+	
SD_CMD_PORT	2	D1−	
		D2+	4 → SD_D0_PORT
		D2−	5 → SD_D1_PORT
		NC	6
		NC	7
		NC	9
		NC	10
		GND	GND
		3V3	GND

U62
TPD4E05U06DQAR

SD_D2_PORT	1	D1+	
SD_D3_PORT	2	D1−	
		D2+	4 → SD_CARD_DETECT
		D2−	5 → GND
		NC	6
		NC	7
		NC	9
		NC	10
		GND	GND
		3V3	GND

3V3_MAIN
C100
22u
GND

3V3_MAIN
R129
47k
SD_CARD_DETECT

3V3_MAIN
C101
100n
GND

CD switch closes to grounded shell on insertion; active LOW. DAT3 is a data contact, not the mechanical detect.

Status and insertion policy

STATUS_LED
R130
1k
LED_A

GND
D1
LED_A
GREEN optional

SD startup allocation 300mA including insertion charging. Card selection, I(V,t), signal integrity and firmware remain OPEN.

Bring-up 400kHz then conservative clock; 4-bit operation test required. Never power a card from an external adapter.

PE6 LED DNP, ~1mA nominal. No new load on 3V3_SENS. No buzzer circuit in this stage.

~~3V3_MAIN~~	3V3_MAIN	
~~GND~~	GND	
~~SDMMC1_CLK~~	SDMMC1_CLK	
~~SDMMC1_CMD~~	SDMMC1_CMD	
~~SDMMC1_D0~~	SDMMC1_D0	
~~SDMMC1_D1~~	SDMMC1_D1	
~~SDMMC1_D2~~	SDMMC1_D2	
~~SDMMC1_D3~~	SDMMC1_D3	
~~SD_CARD_DETECT~~	SD_CARD_DETECT	
~~STATUS_LED~~	STATUS_LED	

RECOMMENDATION — OPEN UNTESTED — NOT FOR FABRICATION

FC Drone Project

Sheet: /Storage and status/
File: 08_storage_status.kicad_sch

Title: FC FC Rev A Stage07 — storage_status

Size: A2	Date: 2026-09-09	**Rev: A-S07-01**
KiCad E.D.A. 10.0.6		Id: 9/9